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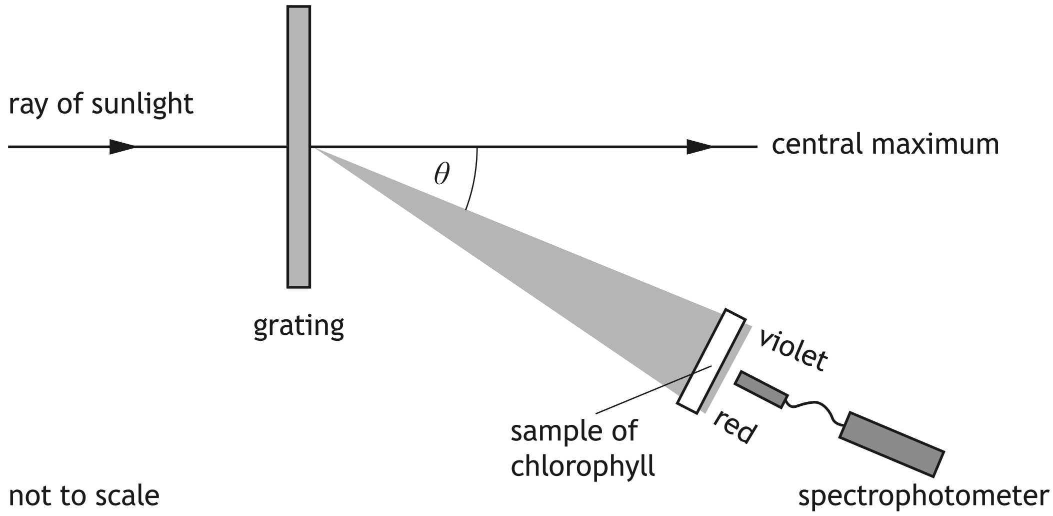
9. A student studies the effect of different wavelengths of light on a sample of chlorophyll.

Chlorophyll is a green pigment, present in all green plants, which is responsible for the absorption of light to provide energy for photosynthesis.

The student sets up the apparatus shown, using a grating with 1000 lines per millimetre to produce an interference pattern.

The interference pattern consists of a central white maximum and a series of spectra on either side of the central maximum.

The student uses the first order spectrum to study the absorption of different wavelengths of light by chlorophyll. The percentage absorption of the different wavelengths of light is measured using a spectrophotometer.



(a) Explain why the central maximum is white. **2**

9. (continued)

(b) The wavelength of light at the violet end of the first order spectrum is 412 nm.

Calculate the angle θ between the central maximum and the violet end of the spectrum. **3**

Space for working and answer

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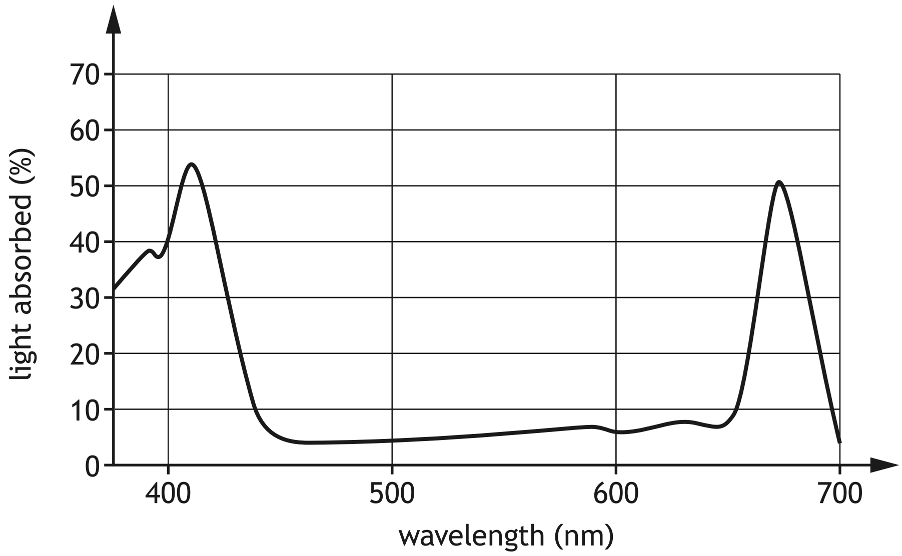
9. (continued)

(c) The graph shows how the percentage of light absorbed by the chlorophyll varies with wavelength of light.



Using information from the graph, suggest why the light passing through the chlorophyll sample appears green. **2**

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9. (continued)

(d) Another student suggests using a grating with 250 lines per millimetre. The student claims that this will allow them to more accurately determine which wavelengths of light are absorbed by the chlorophyll.

Explain why the student's suggestion is incorrect. **2**

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